

Analyzing the Determinants of House Prices and Predicting Sale Prices Using Machine Learning

Arda Ayan

Motivation

The main motivation behind this project was to apply the concepts learned in DSA 210 to a complete data science workflow on a problem that is practical, understandable, and suitable for both statistical analysis and machine learning. House prices were selected as the project topic because they are influenced by many structural and locational characteristics such as living area, construction quality, age, garage capacity, and neighborhood. This makes house price prediction a natural regression problem and a good setting for exploratory data analysis, hypothesis testing, and predictive modeling.

Another reason for choosing this topic was that it allows clear interpretation of both the analysis and the modeling results. Instead of working on a highly abstract problem, I wanted to study a case where the variables are intuitive and the output has direct real-world meaning. The project also gave me the opportunity to practice the full data science pipeline, including data collection, preprocessing, visualization, statistical testing, model building, model comparison, and interpretation of the results.

Data Source

The main dataset used in this project is House Prices - Advanced Regression Techniques, a publicly available Kaggle dataset containing residential property information and sale prices. The training dataset contains 1,460 observations and a large number of explanatory variables describing structural, quality-related, and locational characteristics of houses. These variables include lot area, neighborhood, year built, overall quality, living area, garage information, basement information, and sale price.

A second housing dataset, House Price Prediction Dataset, was also collected from Kaggle during the project planning stage. However, the main analysis and all machine learning stages in the final version of the project were carried out only on the House Prices - Advanced Regression Techniques dataset because it provides a much richer feature set and supports more detailed analysis.

The data was collected by downloading the datasets from Kaggle and importing them into Python for preprocessing, exploratory data analysis, hypothesis testing, and machine learning. In the working version of the main dataset, the Id column was removed because it has no predictive value. Numerical missing values were filled with the median and categorical missing values were filled with the mode. After these steps, the dataset was prepared for both statistical analysis and regression modeling.

Data Analysis

The project was carried out in several stages.

First, I performed an initial inspection of the dataset by examining its shape, variable names, data types, summary statistics, and missing values. This helped me understand the structure of the data before applying any transformations. After removing the Id column, I handled missing values and created several additional features to enrich the analysis. These engineered features included HouseAge, RemodAge, TotalSF, and HasGarage.

Second, I conducted exploratory data analysis to study the distribution of the target variable and its relationship with important housing features. Histograms, scatterplots, box plots, and a correlation heatmap were used to examine variables such as SalePrice, GrLivArea, OverallQual, GarageCars, Neighborhood, HouseAge, and TotalSF. The EDA stage suggested that quality, living area, total square footage, garage-related features, and neighborhood are strongly associated with sale price.

Third, I applied hypothesis testing in order to evaluate whether some of the observed relationships were statistically significant. The following tests were used:

- Independent samples t-test for garage presence and sale price
- One-way ANOVA for overall quality and sale price
- Pearson correlation for above-ground living area and sale price
- One-way ANOVA for selected neighborhoods and sale price

The results of these tests supported the main patterns observed in the visual analysis and showed statistically significant relationships between these housing characteristics and sale price.

Finally, I moved to the machine learning stage. Since SalePrice is a continuous target variable, the problem was treated as a regression task. Before training the models, categorical variables were transformed using one-hot encoding. The dataset was then split into training and test sets using an 80/20 split.

For the machine learning stage, I applied the following models:

- Linear Regression
- Ridge Regression
- Random Forest Regressor

The purpose of adding Ridge Regression was to test whether regularization could improve the baseline linear model. The models were evaluated on the test set using MAE, RMSE, and R^2 .

Findings

This project produced several important findings.

First, the exploratory analysis showed that house prices are strongly associated with a set of structural and quality-related variables. Larger houses, higher-quality houses, and houses with stronger garage-related features generally had higher sale prices. Neighborhood also appeared to play an important role in housing valuation.

Second, the hypothesis tests confirmed that several important relationships are statistically significant. Houses with and without garages differed significantly in sale price, overall quality groups had significantly different mean sale prices, above-ground living area had a strong positive correlation with sale price, and sale prices differed significantly across selected neighborhoods. These results reinforced the visual patterns identified during the EDA stage.

Third, the machine learning models were able to predict sale prices with reasonably strong performance. The model comparison results are summarized below:

The machine learning model results are summarized in Table 1.

Table 1. Regression model performance on the house price dataset.

Model	MAE	RMSE	R ²
Linear Regression	20,157.23	51,289.82	0.6570
Ridge Regression	19,436.96	32,972.28	0.8583
Random Forest Regressor	18,187.92	30,288.60	0.8804

Table 1. Performance comparison of the three regression models.

These results show that Ridge Regression improved the baseline linear model substantially, which suggests that regularization helped stabilize the regression model after one-hot encoding increased the number of input features. However, Random Forest Regressor still achieved the best overall performance on all three evaluation metrics.

Finally, the feature importance analysis from the Random Forest model showed that OverallQual and TotalSF were the most influential predictors, followed by variables such as 2ndFlrSF, YearBuilt, LotArea, and garage-related features. This is consistent with the earlier EDA and hypothesis testing results, where house quality and total size already emerged as major determinants of sale price.

Limitations and Future Work

Although the project produced meaningful results, it also has several limitations.

One limitation is that the analysis mainly relies on a single primary dataset from Kaggle. While the dataset is rich and suitable for a course project, it does not necessarily represent broader housing markets across different countries, time periods, or economic conditions. Future work could benefit from incorporating external information such as neighborhood-level socioeconomic indicators, school quality, interest rates, or local housing market conditions.

Another limitation is that the machine learning models were used with limited hyperparameter tuning. While Ridge Regression and Random Forest Regressor already produced strong results, additional tuning, cross-validation, and more systematic model selection could further improve performance and robustness. More advanced ensemble methods such as Gradient Boosting Regressor or XGBoost-like approaches could also be explored in future work.

A further limitation is that the final project focused primarily on predictive performance rather than model interpretability at a deeper level. Future work could include explainability methods and residual analysis in order to better understand specific prediction errors and the role of individual features in different parts of the housing market.

Conclusion

This project followed the main stages of a data science workflow to analyze and predict house prices. Starting from a real-world housing dataset, I carried out data cleaning, feature engineering, exploratory analysis, hypothesis testing, and machine learning modeling. The results showed that

housing quality, total size, living area, garage-related variables, and neighborhood are strongly associated with sale price.

Among the tested models, Random Forest Regressor achieved the strongest overall performance, while Ridge Regression improved the baseline Linear Regression model through regularization but did not surpass the Random Forest results. Overall, the project helped me apply course concepts in a practical setting and improved my understanding of how data science methods can be used to analyze and predict meaningful real-world outcomes.

References

1. Kaggle. House Prices - Advanced Regression Techniques. Public housing dataset used as the main data source for the project. Available at: <https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques>
2. Kaggle. House Price Prediction Dataset. Secondary housing dataset collected during the planning stage for possible comparison. Available at: <https://www.kaggle.com/datasets/zafarali27/house-price-prediction-dataset>
3. DSA 210 – Linear Regression course slides. Used as background reference for the regression modeling approach and regression framing of house price prediction.
4. DSA 210 – Ensemble Learning course slides. Used as background reference for Random Forest and ensemble-based regression methods.
5. DSA 210 Introduction to Data Science Project Guidelines. Used for final report structure, deliverables, and academic integrity requirements.

AI Assistance Disclosure

AI tools were used in a limited and supportive role during this project. They were mainly used for brainstorming, improving the organization and wording of the report, and getting suggestions on how some analysis steps could be presented more clearly. All coding, execution, output checking, interpretation of results, and final submission decisions were completed by me.